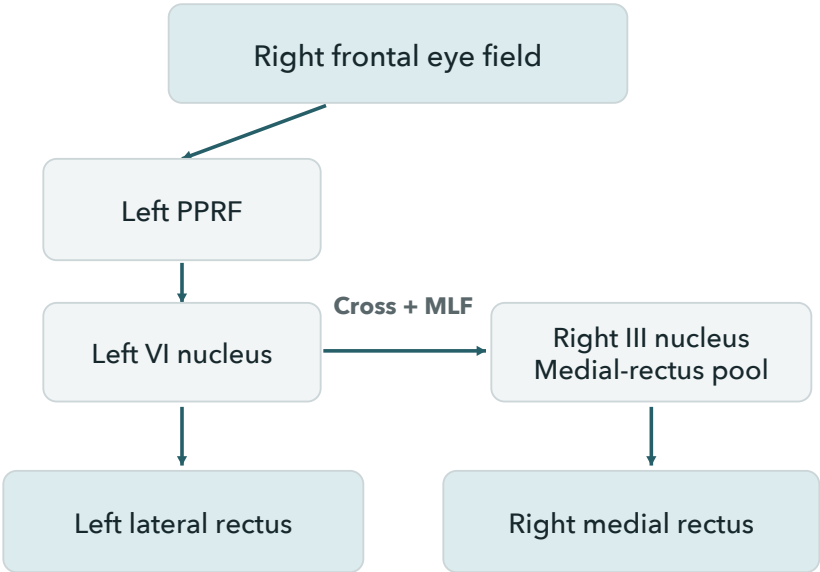


Saccades and the VOR

Redirect fixation or stabilise it: the goal selects the drive.

Voluntary horizontal saccades

A | Right frontal eye field drives left gaze



LEFTWARD CONJUGATE SACCAD

Example follows a leftward command. VI interneurons cross and ascend in the MLF to the opposite medial-rectus motor pool.

Level and expected effect

Level	Consequence
Right frontal eye field	Promotes leftward conjugate saccades
Left PPRF / VI gaze circuitry	Coordinates left lateral rectus with right medial rectus
Superior colliculus	Parallel spatial-to-motor route; important for rapid orienting

Fast check

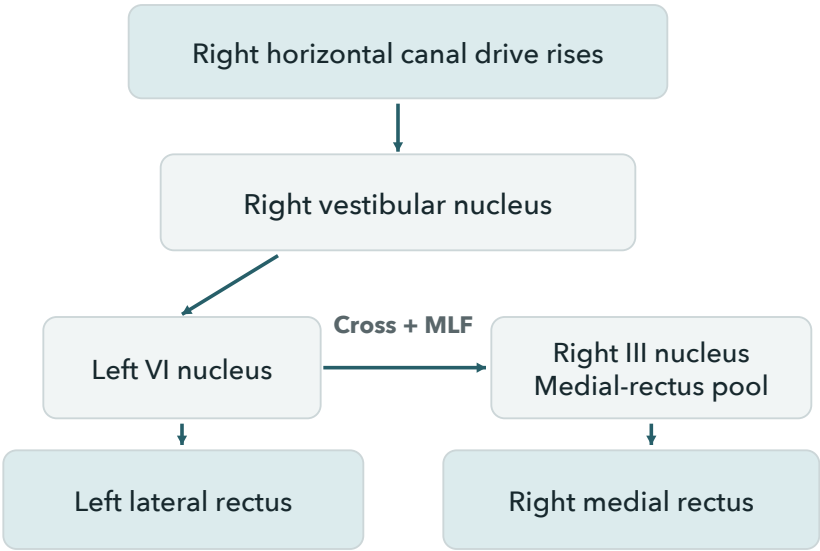
A destructive right frontal eye-field lesion can initially weaken voluntary left gaze and bias the eyes rightward. Preserved reflex gaze helps separate command failure from final muscle paralysis.

Figure reference

A: Neuroscience saccadic eye movements. Original route schematic.

VOR: oppose the head movement

B | Head right, eyes left
HEAD RIGHT → EYES LEFT



The diagram shows the rapid excitatory route. Reciprocal inhibition of antagonist muscles complements it.

Compare the drives

Movement	Trigger	Purpose
Saccade	Selected target / orienting signal	Move fixation quickly
VOR	Head rotation sensed by vestibular apparatus	Stabilise target on retina
Convergence	Near target	Align both foveas for near viewing

Fast check

For a rightward head turn, the left lateral rectus and right medial rectus contract. This compensatory movement can occur before conscious recognition of a new visual target.

Figure reference

B: oculovestibular reflex and eye-movement references.
Original schematic.